

COMPUTER NETWORK

Page No.

Date: / /

Unit-2

ONE SHOT + 3 PYQ SOLUTIONS

Topics :-

- Data Link layer → Services
- Framing and its type
- Error Detection & Correction.
- Error Detection method — 2022-23
- Flow CONTROL
- STOP N WAIT → 10 marks (2021-22, 18-19)
- Sliding window protocol.
- Channel Allocation.
- Multiple Access protocols. → (2021-22, 22-23)
- LAN Standards.
- Bridging & Spanning Tree Algo.

Data Link Layer

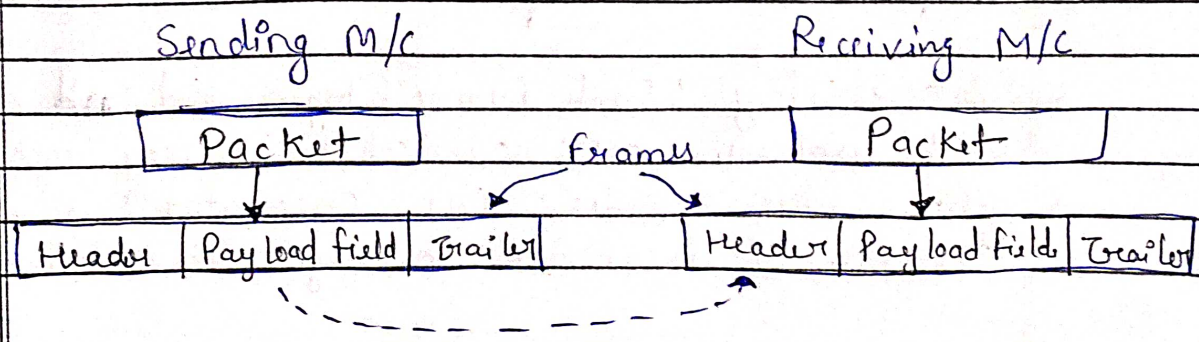
- It is second layer of OSI Layered Model.
- Data link layer is responsible for converting data stream to signals bit by bit and to send that over the underlying hardware.

Services of Data Link Layer

- Framing & Link access
- Reliable Delivery
- Flow Control
- Error Detection
- Error Correction
- Half-Duplex & Full-Duplex

Framing

- Data link layer takes the packets from the Network Layer and encapsulates them into frames. If the frame size becomes too large, then the packet may be divided into small sized frames.
- Smaller sized frames makes flow control and error control more efficient. At receiver's end, data link layer picks up signals from hardware and assembles them into frames.



Parts of a Frame:

- Frame Header: It contains the source and the destination addresses of the frame.
- Payload field: It contains the message to be delivered.
- Trailer: It contains the error detection & error correction bits.
- Flag - It marks the beginning and end of the frame.



Types of Framing

1. Fixed Size

2. Variable size → Length fixed ✓
→ End Delimiter ✓

Character Oriented
Framing (Bytes)
(8bit)

Bit Oriented
Framing.
(1bit)

A Byte stuffing ⇒ Sender's DLL Insert special escape (ESC) Just before 'Accidental data (Byte)'

B Bit Stuffing ⇒ Each frame begin and ~~end~~ end with pattern (01111110) that is flag byte.

→ when even sender's DLL encounter's five consecutive 1s in the message it automatically

stuff '0' bit into outgoing bit stream.

Error Detection and Correction [2022-23]-10marks

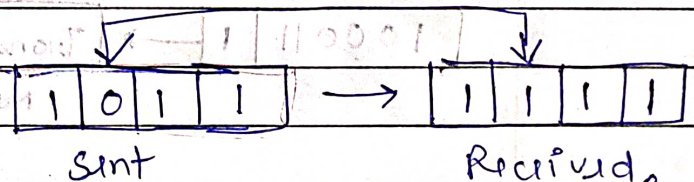
Errors are introduced into binary data transmitted from the sender to the receiver due to noise during transmission.

Error Detection: methods are used to check whether the receiver has received correct data or corrupted data.

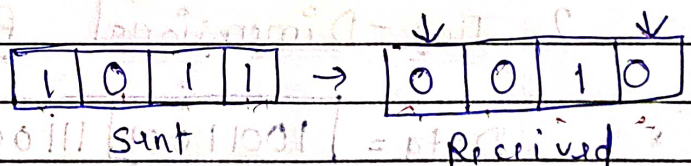
Error Correction: It is used to correct the detected errors during the transmission of data from sender to receiver.

Types of Errors.

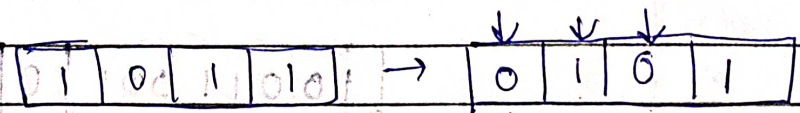
1. Single bit Error →



2. Multiple bit Error →



3. Burst Error →



Error Detection Method

- Simple Parity Check
- 2-Dimensional parity check
- Checksum
- Cyclic Redundancy check

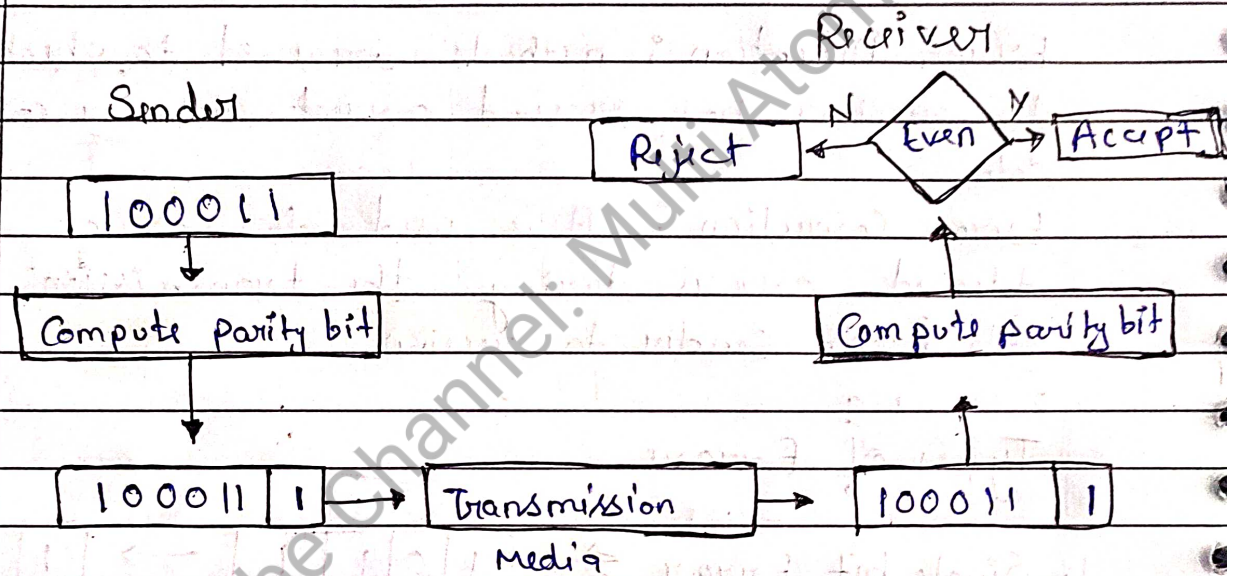
Error Correction

- Type ↓
- Backward EC
- Forward EC
- Techniques ↓
- 1. Hamming Code ✓
- 2. Determining Parity bit

* Even parity $\begin{cases} \rightarrow 1011 \rightarrow 10111 \\ \rightarrow 1010 \rightarrow 10100 \end{cases}$

* Odd Parity $\begin{cases} \rightarrow 1011 \rightarrow 10110 \\ \rightarrow 1010 \rightarrow 10101 \end{cases}$

1. Simple Single Parity Check.



2. Two-Dimensional Parity Check

Data =

10011001	11100010	00100100	10000100
----------	----------	----------	----------

	10011001	0	← row
	11100010	0	
	00100100	0	
	10000100	0	
co ↓	11011011	0	

→

100110010	111000100	001001000	100001000
110110110			

3. Checksum

10011001 11100010 00100100 10000100
1 2 3 4

Sender

Receiver

1 → 10011001

Same as Sender

2 → 11100010

10011101

→ 1

0111100

3 → 00100100

10100000

4 → 10000100

100100100

→ 1

Sum: 00100101

checksum: 11011010

Sum: 00100101

checksum: 11011010

11111111

Complement = 00000000

Conclusion: Accept Data

4. Cyclic Redundancy Check (CRC):

→ eqn → $x^3 + x^2 + 1 = 101101$
 $x^4 + x^2 + 1 = 10101$

$x^4 \ x^3 \ x^2 \ x^1 \ x^0$

→ Frame = 1010000

→ Generator = $x^3 + 1 \Rightarrow 1001$ 4 bit.

→ if CRC generator is of n bit, then append (n-1) zeros in the end of original message.

$X \rightarrow \text{XOR}$

Sender

1001 | 101 0000 000

X = 1001

001 1000 000

X 1001

01010000

X 1001

0011000

X 1001

01010

X 1001

0011

Remainder

→ Message to be transmitted. ⇒ 1010000000

+ 011

1011010000011

Receiver

1001 | 10100000011

X 1001

0011000011

X 1001

01010011

X 1001

0011011

X 1001

01001

X 1001

0000

← Zero means data is accepted.

AKTU - 2022-23 [10 marks]

Page No.:

Date: / /

Q. A bit stream 10011101 is transmitted using x^3+1 generator polynomial. Generate the CRC code word for this message.

$$\begin{array}{r} \rightarrow \quad 1001 \overline{) 10011101000} \\ \underline{\times 1001} \\ 00001101000 \\ \underline{\times 1001} \\ 001000000 \\ \underline{\times 1001} \\ 000100 \leftarrow \text{Remainder!} \\ \underline{\times 1001} \end{array}$$

$\rightarrow$ CRC Code = 1001110100

Hamming Code

→ It is a set of error-correction codes that can be used to detect and correct bit errors that can occur when computer data is moved or stored.

→ It uses parity bit.

→ Using more than one parity bit, an error-correction code is the use of extra not only identify a single bit error in the data unit, but also its location in the data unit.

① → It can be applied to data units of any length.

② → no. of parity bit is decided by

$$2^r \geq m + r + 1$$

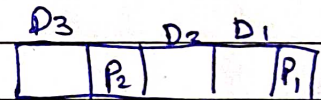
r = No. of parity

m = message bits.

e.g. $m = 4$ [1010]

$$2^3 \geq 4 + 3 + 1$$

$$8 \geq 8 \checkmark$$

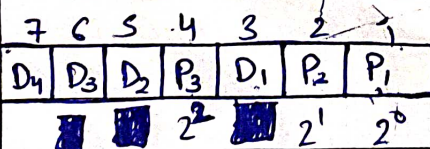
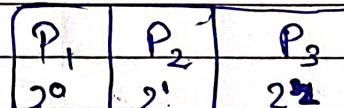


③ → This parity bits are positional parity bits

$m = 4$ bit

$r = 3$ bit

$4 + 3 = 7$ bit



E.g. A bit word 1011 is to be transmitted construct the even parity seven bit Hamming code for this data.

→ even parity = No. of 1's will be Even. ✓

Sol $m = 4 \text{ bit}$
 $r = 3 \text{ bit}$

→ $2^r \geq m + r + 1$

step 1 $\Rightarrow r = 0 = [2^0 \geq 4 + 0 + 1] \times$

$1 = [2^1 \geq 4 + 1 + 1] \times$

$[2^2 \geq 4 + 2 + 1] \times$

$[2^3 \geq 4 + 3 + 1] \checkmark$

$r = 3$

$P_1 \quad P_2 \quad P_3$
 $2^0 \quad 2^1 \quad 2^2$

→

7	6	5	4	3	2	1
D_4	D_3	D_2	P_3	D_1	P_2	P_1

→

7	6	5	4	3	2	1
1	0	1	P_3	1	P_2	P_1

111 110 101 100 011 010 001

$P = 3$ Data

$P_1 = (3, 5, 7) \rightarrow (1, 1, 1) = 1$

$P_2 = (3, 6, 7) \rightarrow (1, 0, 1) = 0$

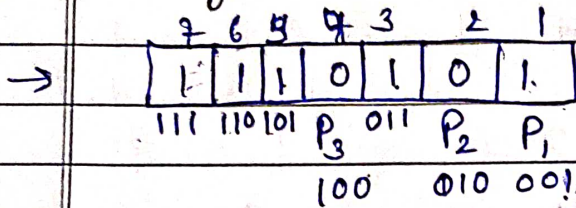
$P_3 = (5, 6, 7) \rightarrow (1, 0, 1) = 0$

⇒

1	0	1	0	1	0	1
---	---	---	---	---	---	---

AKTU-2022-23 [2 marks]

Q. If a 7-bit hamming code received as 1110101, show that the code word has error. Also, rectify error in this code.

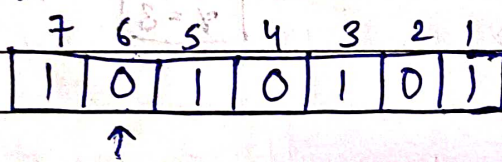


$$P_1 = 1, 3, 5, 7 = (0, 1, 1, 1) = 0$$

$$P_2 = 2, 3, 6, 7 = (0, 1, 1, 1) = 1$$

$$P_3 = 4, 5, 6, 7 = (0, 1, 1, 1) = 1$$

$$0110 = 6$$



Flow Control

Feedback based

→ [Sender gets ack. from the user]

Rate Based

→ [check the rate of flow without ack.]

Protocols

for

Noiseless Channels

→ Simplex protocol

→ stop & wait protocol

for

Noisy channels

→ stop & wait ARQ

→ Go-Back-N ARQ

→ selective Repeat ARQ

① ~~Elementary~~ AKTU - 2022-23 [10 marks]

Q Explain error control ~~control~~ mechanism in Data Link layer and giving example of each method

Ans ① Elementary Data Link Protocol → simplex & stop and wait

② Sliding Window Protocols → Go-Back-NARQ
→ Selective Repeat ARQ

① Simplex Protocol

→ As the name suggests, it is the most basic Data Link protocol.

→ data can only be sent in a single direction.

→ the Sender/Receiver can generate an infinite amount of data.

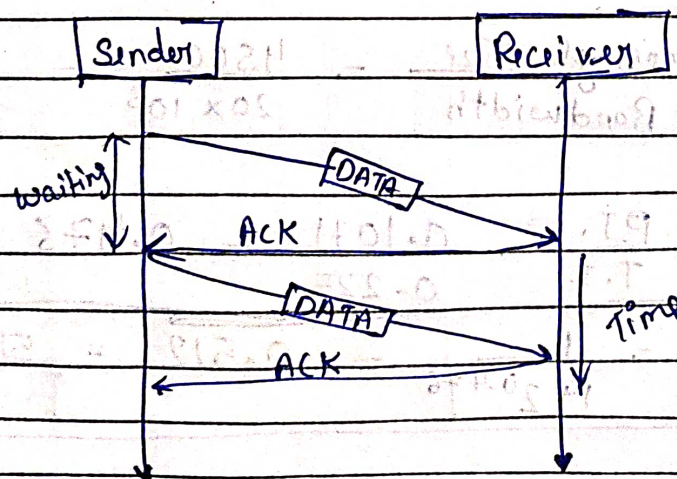
→ e.g → Keyboard transmitting keystrokes to a Computer

② Stop and Wait Protocol

→ It is a fundamental data link protocol where the sender transmits a frame and waits for an ack.

from the receiver before sending the next frame.

→



AKTU-2018-19 [10 marks]

Page No.

Date / /

Q. A channel has a bit rate of 20Kbps. The stop and wait protocol with frame size 4500 bit is used. The delay for error detection and sending ACK by the receiver is 0.25 seconds because of a fault. Find the maximum efficiency of the channel if destination is 30000 km away and the speed of the propagation of the signal is 2.8×10^8 m/s. Find the decrease in efficiency due to the fault.

Sol Bandwidth = 20×10^3 bps
 Size = 4500 bit
 distance = $30,000 \times 10^3$ m
 propagation speed = 2.8×10^8 m/s
 delay because of fault = 0.25 sec
 decrease in efficiency = ? & Max efficiency = ?

$$\rightarrow \eta_{\max} = \frac{1}{1 + 2a} \quad a = \frac{\text{Propagation Time}}{\text{Transmission Time}}$$

$$\rightarrow P.T = \frac{\text{Distance}}{\text{Propagation Speed}} = \frac{30,000 \times 10^3}{2.8 \times 10^8} = 0.1071$$

$$T.T = \frac{\text{Message Size}}{\text{Bandwidth}} = \frac{4500}{20 \times 10^3} = 0.225$$

$$a = \frac{P.T.}{T.T.} = \frac{0.1071}{0.225} = 0.476$$

$$\eta_{\max} = \frac{1}{1 + 2 \times 0.476} = \frac{1}{1.952} = 0.512 = 51.2\%$$

$$\eta_{\text{fault}} = \frac{1}{1 + 2a + \frac{0.25}{T.T.}} = \frac{1}{1 + 2 \times 0.476 \times \frac{0.25}{0.225}}$$

$$\eta_{\text{fault}} = 0.326 = 32.6\%$$

$$\rightarrow \text{decrease in } \eta (\%) = \frac{\eta_{\text{max}} - \eta_{\text{fault}}}{\eta_{\text{max}}} \times 100$$

$$= \frac{0.512 - 0.326}{0.512} \times 100$$

$$= 36.3\%$$

AKTU- 2021-22 [10 marks]

Q. Define the relationship b/w Transmission delay & Propagation delay, if the efficiency is at least 50% in STOP n wait protocol.

$$\rightarrow \eta = \frac{1}{1 + 2a}, \quad a = \frac{P.T.}{T.T.}, \quad \eta = 0.50 = \frac{1}{2}$$

$$= \frac{1}{2} = \frac{1}{1 + 2 \times \frac{P.T.}{T.T.}}$$

$$= 1 + 2 \frac{P.T.}{T.T.} = 2$$

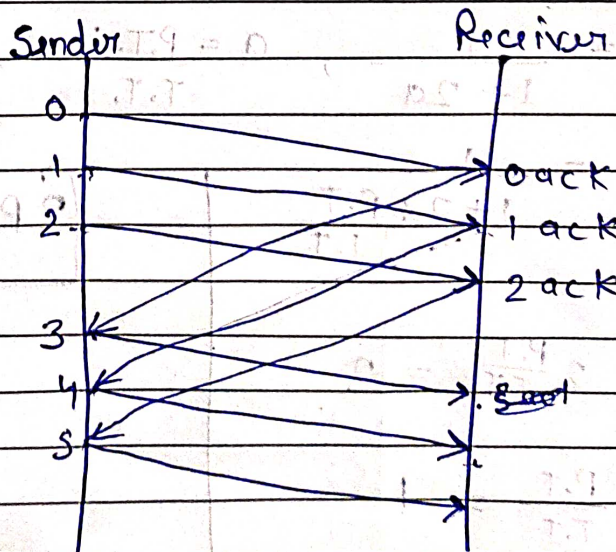
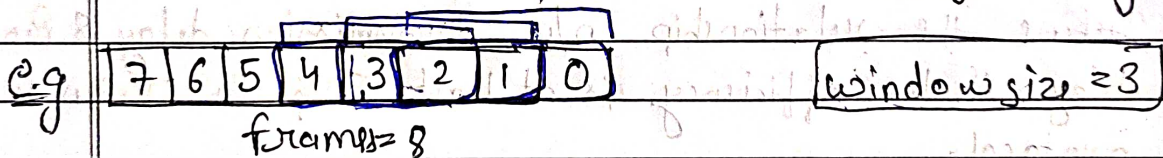
$$= 2 \times \frac{P.T.}{T.T.} = 1$$

$$2 \text{ Propagation Time} = \text{Transmission Time.}$$

→ Sliding window Protocol

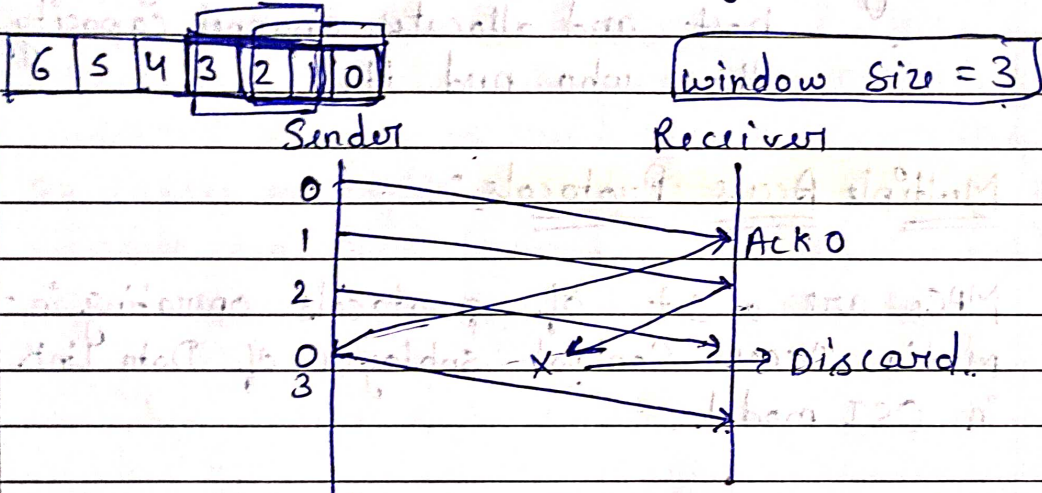
- * It is a technique for controlling transmitted multiple data packets b/w two network computers.
- * Each data packet includes a unique consecutive Sequence number, which is used by Receiver
- * to place data in the correct order.
- avoid duplicate data & to request missing data.
- It uses the concept of piggy backing - Instead of sending ack frame on its own, if there is an outgoing data frame in the next short interval, attach the ack to it (using "ack" field in header).

AKTU - 2022-23 [2 mark] - (piggy backing.)



① GO-BACK-N ARQ (Automatic Repeat Request)

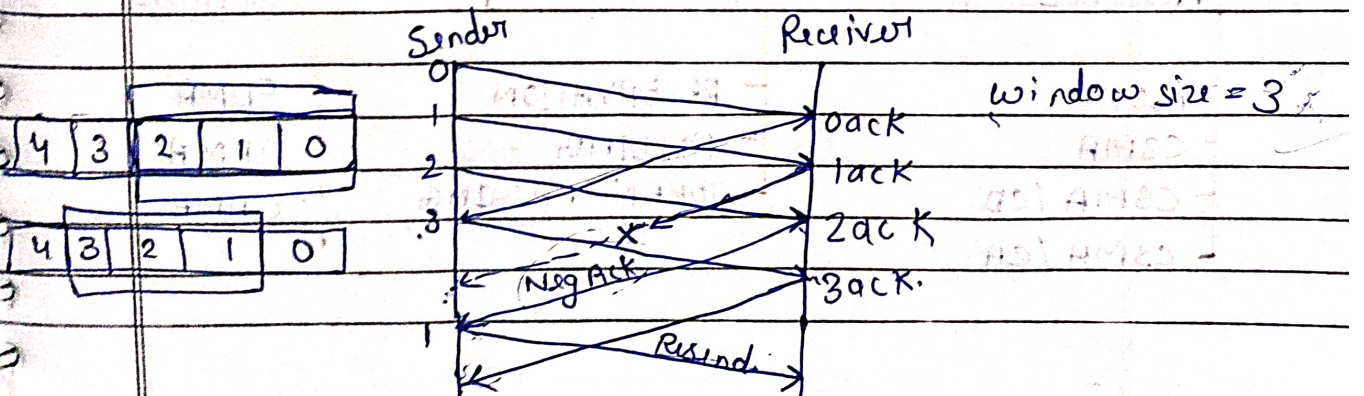
- If one frame is corrupted then all frames have to be sent again. (as respect window size)
- Number of frames sent depends on window size.
- The size of the sender window is N and the receiver window size is always 1.



- Now, Sender have to sent 1, 2, 3 again to Receiver

② SELECTIVE REPEAT ARQ [Automatic Repeat Request].

- In this only the frame is sent again, which is lost.
- Size of sender window = size of Receiver window.
- Size of the sliding window is always greater than 1.
- when receiver receives corrupt frame. It sends a Nig ACK.



Channel Allocation

→ It is a process in which a single channel is divided and allotted to multiple users in order to carry user specific tasks.

polling → A central controller interrogates each host and allocates channel capacity to those who need it

Multiple Access Protocols:

MACs are a set of protocols operating in the Media Access Control sublayer of Data Link Layer in OSI model.

MULTIPLE ACCESS PROTOCOLS

RANDOM ACCESS PROTOCOL

- ALOHA
- CSMA
- CSMA/CD
- CSMA/CA

CONTROLLED ACCESS PROTOCOL

- RESERVATION
- POLLING
- TOKEN PASSING

CHANNELIZATION PROTOCOL

- FDMA
- TDMA
- CDMA

1. **Random Access Protocols** → Assign uniform priority to all connected nodes.

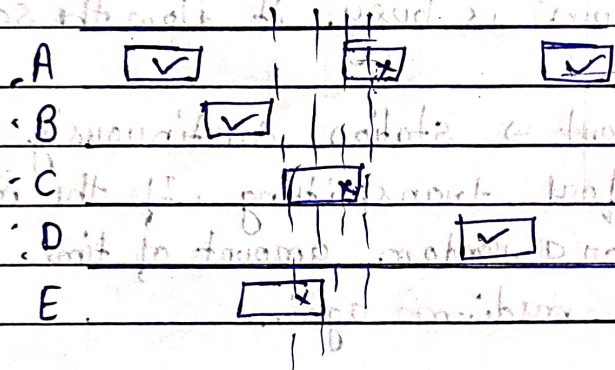
→

ALOHA

Pure ALOHA

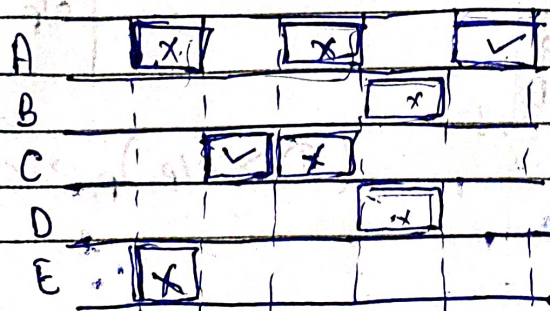
Slotted ALOHA

1. **PURE ALOHA** ⇒ Devices transmit data whenever they have it, without checking if the channel is busy. Collisions are detected through ack. messages or by the sender if no ack. is received after a certain time.
- The total vulnerable time of pure ALOHA is $2 \times T_{fr}$
- effective maximum channel utilization is 18.4%

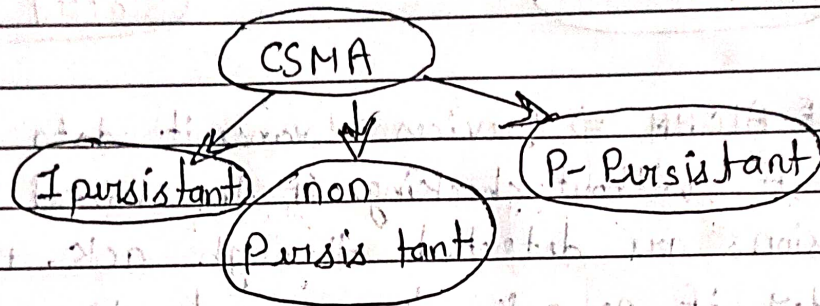


2. **Slotted ALOHA** ⇒ This variant divides time into discrete slots, and devices are only allowed to transmit at the beginning of each slot. This reduces the probability of collisions compared to pure ALOHA. However, collisions can still occur.

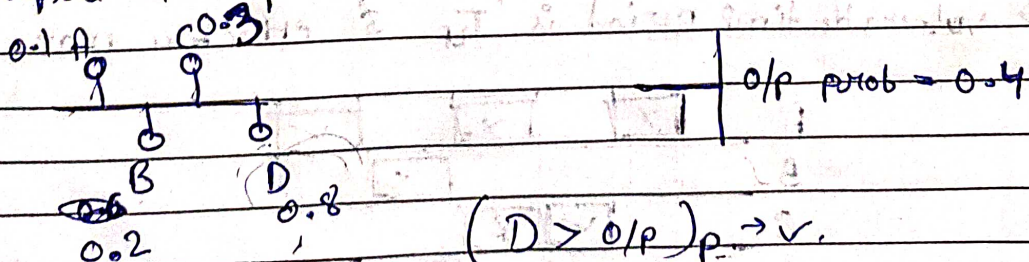
→ vulnerable time period is T_{fr} & effective max. utilization → 36.8%



- CSMA (Carrier Sense Multiple Access).
- devices listen to the communication channel before transmitting. If the channel is idle, the device can proceed with transmission and if the channel is busy the device will wait.



- 1-persistent → Continuously senses the medium. If the medium is idle, it immediately begins transmission. If the medium is busy, it follows the same process.
- Non-persistent → station continuously senses the medium before transmitting. If the medium is busy, it waits for a random amount of time and then checks the medium again.
- p-persistent → when a station wants to transmit data, it senses the medium. If the medium is idle, it checks if (device probability > output prob.). if true, it transmits; otherwise, it waits for a slot and repeats the process.



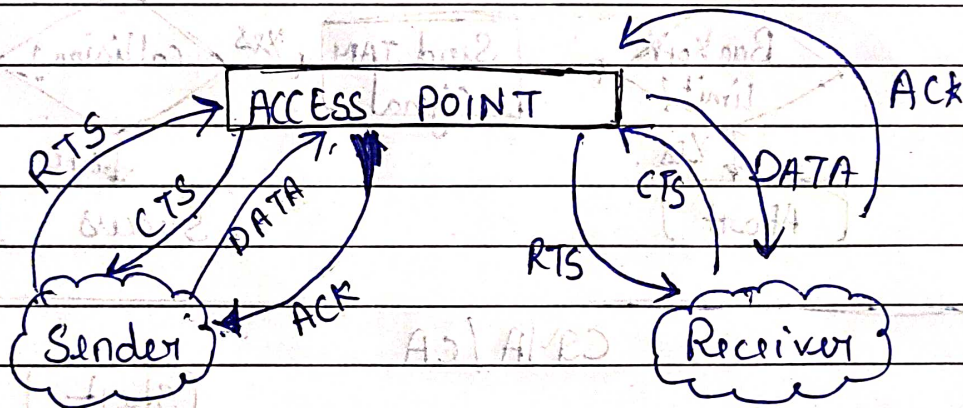
AKTU-2021-22 + Flowchart

→ CSMA/CD (Carrier Sense Multiple Access with Collision Detection).

* Commonly used in Ethernet networks, not only listens to the channel before transmitting but also detects collision while transmitting. If a collision is detected, the node stop transmission and send JAM signal to all other nodes and waits for a random backoff time.

→ CSMA/CA (Carrier Sense Multiple ^{Access} with Collision Avoidance)

* used in wireless networks like WiFi.



RTS → Request to send

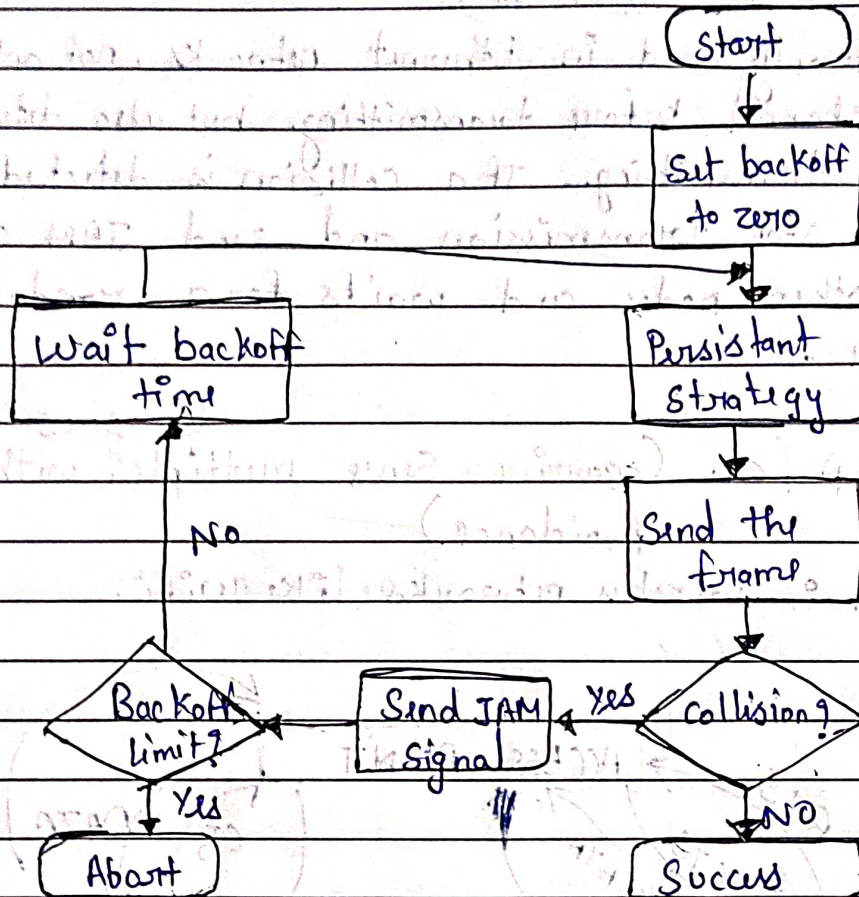
CTS → clear to send

AKTU - 2022-23 (10 marks)

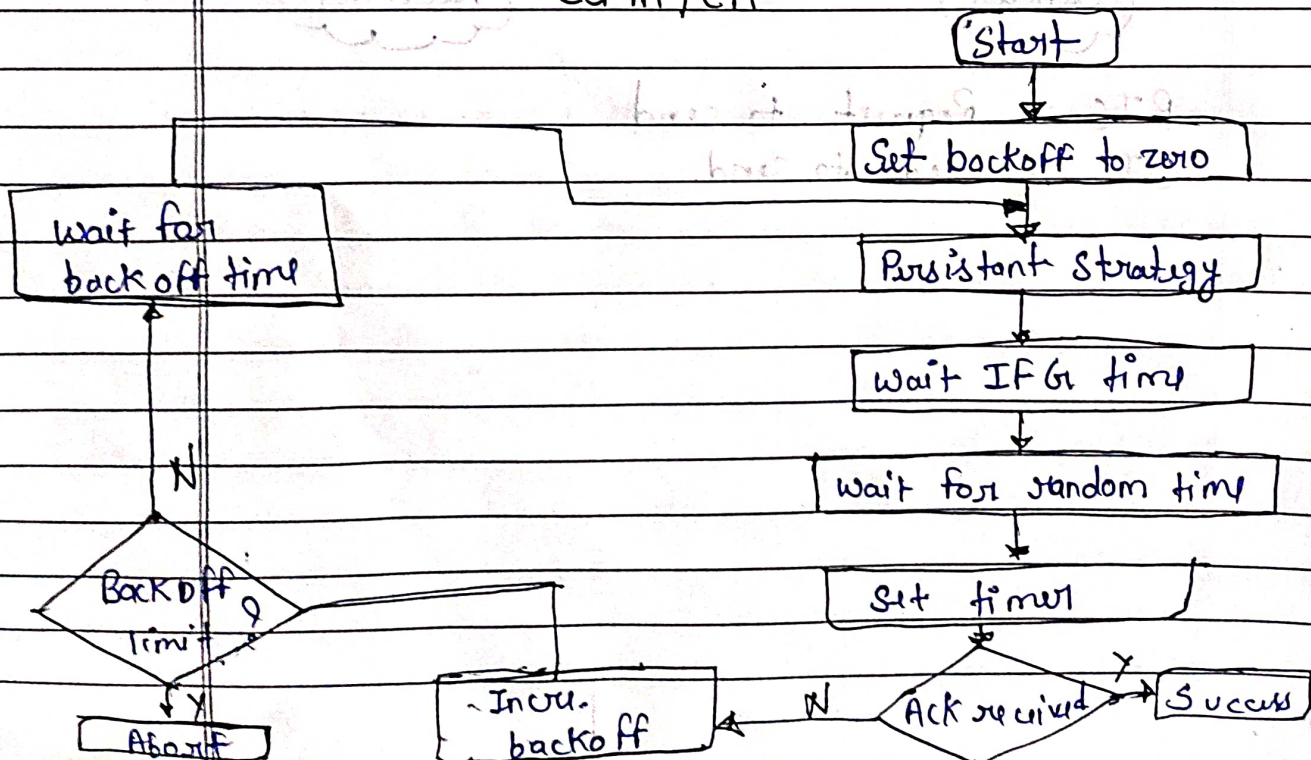
Page No. .

Date. / /

CSMA / CD



CSMA / CA



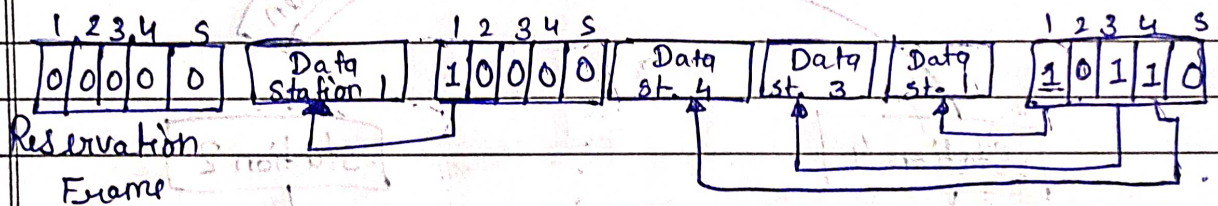
2. Controlled Access Protocol

→ It allows only one node to send at a time, to avoid collision of messages on shared medium

A. Reservation

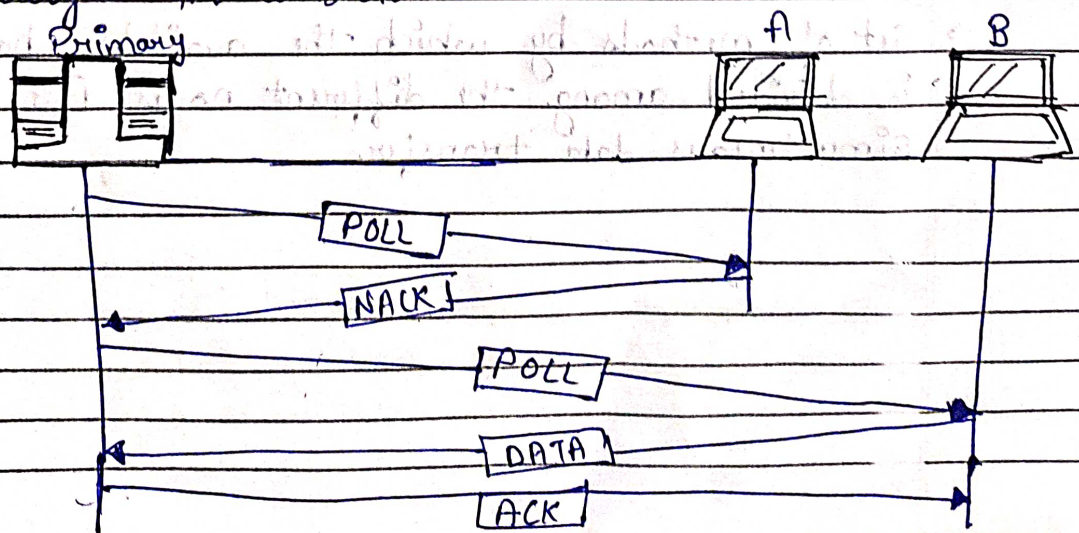
→ A station needs to make a reservation before sending the data

→ whenever a station needs to send the data frame, then the station makes a reservation in its own minislot.



B. Polling

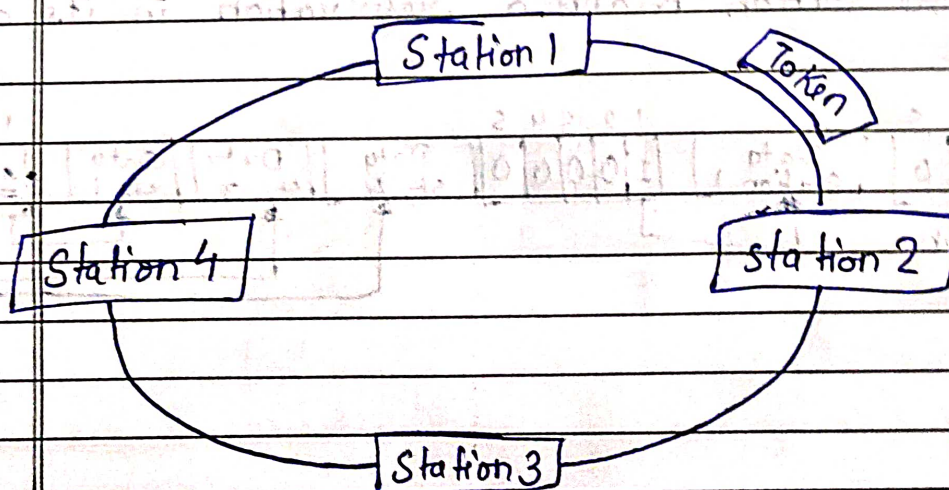
→ In this, one acts as a controller and the others are secondary stations. All data exchanges must be made through the controller.



C. Token Passing

→ the stations are connected logically to each other in form of ring and access to stations is governed by tokens. (short message)

→ drawbacks are duplication of token or token is lost or insertion of new station, removal of a station.

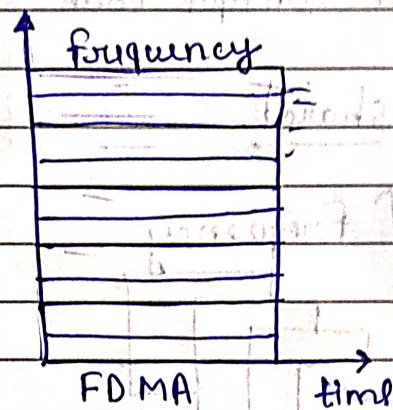


3. Channelization Protocol

→ Set of methods by which the available bandwidth is divided among the different nodes for simultaneous data transfer.

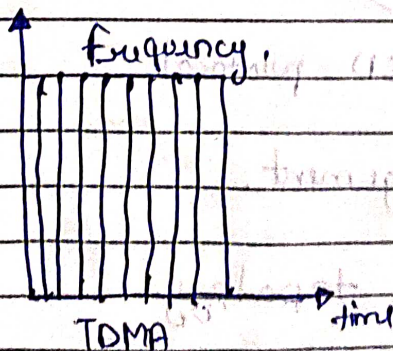
A FDMA (Frequency Division Multiple Access)

→ In this bandwidth is divided into various frequency bands. Each station is allocated with band to send data and that band is reserved for particular station for all the time, which is as follows:



B TDMA (Time Division Multiple Access)

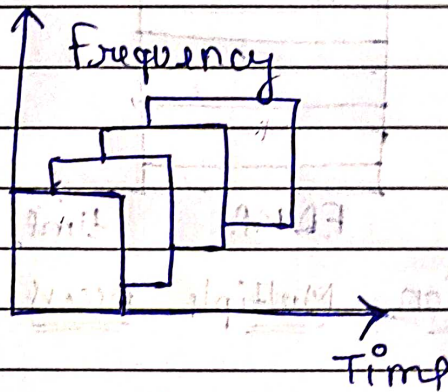
→ TDMA is the channelization protocol in which bandwidth of channel is divided into various stations on the time basis. There is a time slot given to each station, that station can transmit data during that time slot only which is as follows:



3. Code Division Multiple Access (CDMA):

- all the stations can transmit data simultaneously. It allows each station to transmit data over the entire frequency all the time. & they are separated by unique code sequence. Each user is assigned with a unique code sequence.

Data + Code → shared.



LAN Standards

* 802.3 frame format

- used by LAN ✓
- use CSMA/CD protocol ✓
- No. acknowledgement ✓
- Bus and star topology ✓

46 to 1500

7 byte	1 byte	6 byte	6 byte	2 byte	↑ byte	4 byte
Preamble	Start Frame Delimiter	Destination Address	Source Address	Length	Data	Frame check Sequence

1. Preamble → 56 bit pattern of 1s or 0s used for Synchronization and timing recovering by the receiving device.
2. SFD → indicating the start of the frame.
3. D.A → indicating the MAC Address of the receiver device.
4. S.A → indicating the MAC Address of the sender device.
5. Length → indicating the type of protocol (e.g. IPv4, ARP) (length of data)
6. Data → Data.
7. FCS → for error detection.

→ Data Link layer Max Data length → 1518-byte. → First 2 marks A.K.T.C

② 802.5 Format

- LAN
- Use for Ring Topology
- Access Control Method for token passing. ✓
- Piggy backing (Data + Ack.)
- Use Differential Manchester encoding.

[802.5] Frame format.

Fiber Distributed Data Interface (FDDI) .
Frame format.

Start Delimiter	Access Control	Frame Control	Dest Add.	Src Add.	Data	FCS	End Deli.	Frame Status
1 byte	1	1	6	6	≥ 0	4	1	1

Start Deli.	Access Control	End Deli.	max. 4500 bytes
1	1	1	

← token format

→ Switching in unit-1 ✓

→ Bridging Concept

Bridge is used to connect two different LAN's



Bridge works at 'physical layer' as well as at "DLL"

Mapping table for bridge.

MAC	A	B	C	D	E	F	G	H
	↓	↓	↓	↓	↓	↓	↓	↓
PORT	P1				P2			

① Static Bridge : Do ~~mapping~~ manually mapping.

② Dynamic / learning / Transparent Bridge :

They will automate learning the entries.

Capabilities of a Bridge :

① Filtering : (S-MAC & D-MAC are on same LAN)

② Forwarding : (S-MAC & D-MAC are on diff. LAN)

③ Flooding : (If any new station will be added)

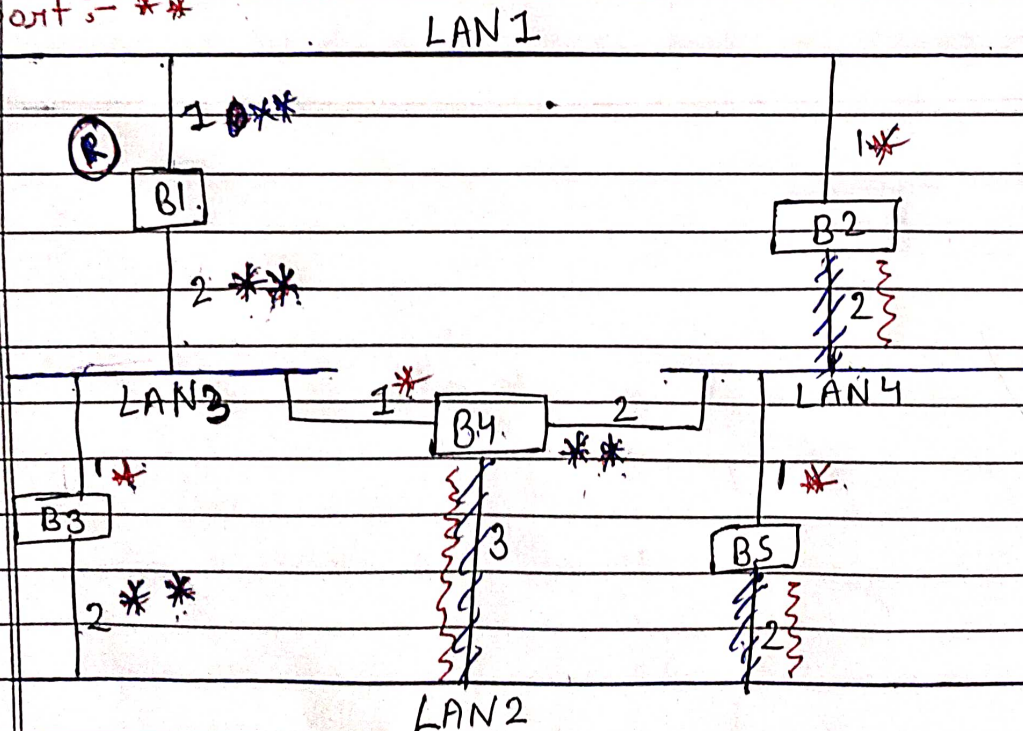
④ Store & forward : (No collision will occur)

Spanning Tree Algorithm for Bridging

- ① Every Bridge is having a built in ID. The one with smallest ID is taken as root bridge.
- ② Mark one port of each bridge which is closest to the root bridge as a root port.
- ③ ~~Each~~ Every LAN chooses a bridge closest to it as a designated bridge for that LAN & marks that correspondant port as a designated port.
- ④ Mark the root port and designated port as forwarding port & block the remaining port.

Root Port :- *

Designated port :- **



Page No.:

Date: / /

Unit-2 is Completed

Subscribe ✓

MULTI ATOMS

Join

TELEGRAM